For an AI Blog Generator Web Application, the **Software Requirements Specification (SRS)** should be written professionally as if it were being submitted for a college project, internship project, or client proposal.

Software Requirements Specification (SRS)

1. Project Title

AI Blog Generator Web Application

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2. Introduction

2.1 Purpose

The purpose of this project is to develop a web-based AI Blog Generator that allows users to generate high-quality blog content automatically using Artificial Intelligence. The application will leverage the GROQ API for content generation and the MERN Stack for development.

The system will help content creators, marketers, students, and businesses create blog articles quickly without spending hours writing manually.

2.2 Scope

The AI Blog Generator will:

- Generate blogs using AI prompts
- Allow users to specify blog title and topic
- Generate SEO-friendly content
- Store generated blogs
- Allow users to edit generated content
- Export blogs
- Maintain user accounts
- Provide blog history

Deployment will be done using GitHub and hosted on Render or Vercel.

2.3 Technologies Used

Frontend

- React.js

- Tailwind CSS
- Axios

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

AI Integration

- GROQ API

Deployment

- GitHub
- Vercel (Frontend)
- Render (Backend)

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3. Overall Description

3.1 Product Perspective

The system is a cloud-based web application that interacts with:

- Users
- MongoDB Database
- GROQ AI API

Architecture

```
User
 ↓
React Frontend
 ↓
Express Backend
 ↓
GROQ API
 ↓
MongoDB Atlas
```

3.2 User Classes

Guest User

Can:

- Visit landing page
- View features
- Register account

Registered User

Can:

- Login
- Generate blogs
- Save blogs
- Edit blogs
- Delete blogs
- Export blogs

Admin (Optional)

Can:

- Manage users
- Monitor blog generation usage
- View analytics

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4. Functional Requirements

FR-1 User Registration

Description

Users should be able to create an account.

Inputs

- Name
- Email
- Password

Outputs

- Account created successfully

FR-2 User Login

Description

Users can authenticate using email and password.

Inputs

- Email
- Password

Outputs

- JWT Token
- Login Success Message

FR-3 Blog Generation

Description

Users can generate blogs using AI.

Inputs

- Blog Title
- Topic
- Keywords
- Tone

Process

- Request sent to backend
- Backend calls GROQ API
- AI generates blog

Output

- Complete blog article

FR-4 Blog History

Description

Generated blogs should be stored.

Inputs

- Blog content
- User ID

Outputs

- Blog saved in MongoDB

FR-5 Edit Blog

Description

Users can edit generated blogs.

Inputs

- Updated content

Outputs

- Updated blog saved

FR-6 Delete Blog

Description

Users can delete blogs.

Outputs

- Blog removed successfully

FR-7 Export Blog

Description

Users can download blogs.

Formats

- TXT
- PDF

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5. Non-Functional Requirements

Performance

- Blog generation response within 5-10 seconds

- Support 100+ concurrent users

Security

- JWT Authentication
- Password Hashing using bcrypt
- HTTPS Communication
- Secure API Keys

Reliability

- 99% uptime
- Automatic error handling

Scalability

- Cloud deployment
- MongoDB Atlas support

Usability

- Responsive design
- Mobile-friendly UI
- Easy navigation

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6. Database Design

Users Collection

JSON

```json
{
  "_id": "",
  "name": "",
  "email": "",
  "password": "",
  "createdAt": ""
}
```

Blogs Collection

JSON

```json
{
  "_id": "",
  "userId": "",
  "title": "",
  "topic": "",
  "content": "",
  "createdAt": ""
}
```

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7. API Requirements

Authentication APIs

Register

http

```http
POST /api/auth/register
```

Login

http

```http
POST /api/auth/login
```

Blog APIs

Generate Blog

http

```http
POST /api/blog/generate
```

Save Blog

http

```
POST /api/blog/save
```

Get Blogs

```http
GET /api/blogs
```

Update Blog

```http
PUT /api/blog/:id
```

Delete Blog

```http
DELETE /api/blog/:id
```

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8. User Interface Requirements

Pages

Landing Page

- Hero Section
- Features
- CTA Button

Login Page

- Email
- Password

Register Page

- Name
- Email
- Password

Dashboard

- Blog Form
- Generated Content

- History Sidebar

Profile Page

- User Information
- Generated Blog Count

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9. Deployment Requirements

Version Control

- GitHub Repository

Frontend Hosting

- Vercel

Backend Hosting

- Render

Database Hosting

- MongoDB Atlas

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10. Future Enhancements

- Multi-language blog generation
- SEO score checker
- AI image generation
- Blog publishing to WordPress
- Content plagiarism checker
- Blog templates
- Team collaboration

Suggested Project Modules (Resume/Interview Friendly)

1. User Authentication Module
2. AI Blog Generation Module
3. Blog Management Module
4. Export PDF Module
5. User Dashboard Module
6. API Integration Module

7. Deployment & CI/CD Module

This scope is appropriate for a **final-year B.Tech project**, internship project, or portfolio project and can realistically be completed using **MERN + GROQ API + MongoDB Atlas + GitHub + Vercel/Render** in 2–4 weeks.